Composite Multi-Objective Bayesian Optimization: Benchmark Suite — Formulations and Results

tau315/composite-mobo

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1 Setup and notation

Each benchmark is a deterministic multi-objective minimization problem with a *composite* structure

$$x \;\longrightarrow\; g(x) \in \mathbb{R}^K \;\longrightarrow\; f(x) = L\big(g(x)\big) \in \mathbb{R}^M,$$

where g is a *raw response* that a real simulator/solver computes and L is a *known*, closed-form reduction. The **composite** variant of a solver fits K surrogates to g and applies L analytically; the **direct** variant fits M surrogates to f. Inputs are normalized to $[0,1]^d$ and mapped to each problem's native bounds.

Solvers. `standard_mobo` (qEHVI), `chebyshev_bo` (augmented-Chebyshev scalarization), `spherical_chebyshev_` (spherical scalarization, high-dim), `batched_morbo` (trust-region MORBO); each in a direct and composite variant. Reported Δ is the composite-minus-direct percentage change in final dominated hypervolume, paired per seed at n seeds ($*$: $p < 0.05$, paired t-test).

2 Synthetic composite benchmarks

2.1 DTLZ2 (M=2; $d \in \{6, 100, 600\}$)

With one position variable x_0 and distance variables $x_1, \ldots, x_{d-1}$,

$$g = \sum_{j=1}^{d-1} (x_j - 0.5)^2, \qquad \theta = \tfrac{\pi}{2} x_0, \qquad f_1 = (1+g)\cos\theta, \quad f_2 = (1+g)\sin\theta.$$

Composite: $g(x)$ exposes the per-variable terms $(x_j - 0.5)^2$ (each a 1-D function); L sums them to g and applies the trig map. **Result** (`batched_morbo`): d=6 +5.3%*, d=100 +0.3%*, d=600 +0.1%. Also M=5, d=6: +0.5%*.

2.2 Ackley–Griewank (M=2; $d \in \{6, 50\}$)

On rotated/shifted coordinates $z = R(x - c)$, the two objectives are a normalized Ackley and Griewank. Ackley uses the two intermediates $\big(\overline{z^2},\ \overline{\cos 2\pi z}\big)$ with

$$\text{Ackley} = -20\exp\big(-0.2\sqrt{\overline{z^2}}\big) - \exp\big(\overline{\cos 2\pi z}\big) + 20 + e;$$

Griewank uses $\big(\frac{1}{4000}\sum z_j^2,\ \prod_j \cos(z_j/\sqrt{j})\big)$ with Griewank $= 1 + \frac{1}{4000}\sum z_j^2 - \prod_j \cos(z_j/\sqrt{j})$. **Composite:** $g(x)$ exposes these mean/sum/product intermediates; L is the outer nonlinearity. **Result:** d=6 +2.7%* (`standard`); d=50 +0.1% (`batched_morbo`).

2.3 Five-Ackley (M=5, d=6) and Langermann–Ackley (M=2, d=6)

Multi-objective stacks of the Ackley intermediates above and of the generalized Langermann map $-\sum_i c_i \exp(-d_i/\pi)\cos(\pi d_i)$ (with d_i squared distances to targets). **Composite:** the per-objective additive terms are the raw response. **Result** (`batched_morbo`): Five-Ackley +5.0%*; Langermann–Ackley +5.1%* (and +8.0%* on `standard`).

2.4 Projected Langermann (M=2, d=500)

A 500-D Langermann whose raw response exposes the low-dimensional projections feeding the exponential-cosine map. **Result** (`spherical_chebyshev`): +0.5%*.

3 Real-world benchmarks

3.1 Summit SnAr plug-flow reactor (M=2, d=4)

Inputs: residence time τ, pyrrolidine equivalents ρ, inlet concentration C_0, temperature T. The five species $C = (C_{\text{dfnb}}, C_{\text{pldn}}, C_{\text{prod}}, C_{\text{regio}}, C_{\text{bis}})$ evolve under Arrhenius kinetics $k_j = 0.6\, k_j^{\text{ref}} \exp[-\frac{E_{a,j}}{R}(\frac{1}{T_K} - \frac{1}{T_{\text{ref}}})]$:

$$\begin{aligned}
\dot C_0 &= -(k_a{+}k_b)C_0C_1, & \dot C_2 &= k_aC_0C_1 - k_cC_1C_2,\\
\dot C_1 &= -(k_a{+}k_b)C_0C_1 - k_cC_1C_2 - k_dC_1C_3, & \dot C_3 &= k_aC_0C_1 - k_dC_1C_3,\\
\dot C_4 &= k_cC_1C_2 + k_dC_1C_3,
\end{aligned}$$

integrated over $[0,\tau]$ with flow $q = V/\tau$ (V=5 mL). Objectives:

$$\text{STY} = \tfrac{6\times10^4}{1000}\,\tfrac{M_2\,C_{\text{prod}}\,q}{V}\ \text{(max)}, \qquad \text{E} = \frac{q\rho_{\text{EtOH}} + 10^{-3}q\sum_{i\neq 2} M_iC_i}{10^{-3}M_2C_{\text{prod}}q}\ \text{(min)}.$$

Composite: $g(x) = (C,\, F_{\text{prod}})$ with product molar flow $F_{\text{prod}} = C_{\text{prod}}q$; STY reads F_{prod} and q cancels in E (leaving a function of C only). **Result:** +0.3%* (`batched_morbo`) — essentially tied (a coupled 4-D reactor ODE).

3.2 Nanoparticle RGB structural color (M=3, d=6)

Inputs: core radius and five shell thicknesses. Mie scattering is evaluated at 201 visible wavelengths, giving a scattering spectrum $S(\lambda; x)$. Each color objective $m \in \{R, G, B\}$ is composed from its in-band and out-of-band integrated intensities,

$$I_m^{\text{in}} = \int_{\text{band } m} S\, d\lambda, \quad I_m^{\text{out}} = \int_{\text{out } m} S\, d\lambda, \qquad f_m = L\big(I_m^{\text{in}}, I_m^{\text{out}}\big).$$

Composite: $g(x) = (I_m^{\text{in}}, I_m^{\text{out}})_m$ (the band integrals); L forms the selectivity objectives. **Result:** +0.2% (`standard`) — tied (a coupled global Mie solve).

3.3 CORT TG119 radiotherapy (M=3, d=418)

418 beamlet intensities $x \geq 0$; public dose-influence matrices D_s per structure s give the voxel dose $\mathbf{d}_s = D_s x$. Objectives are target-coverage / organ-sparing functionals of the per-voxel doses. **Composite:** $g(x)$ exposes the per-structure dose statistics; L forms the three clinical objectives. **Result:** +1.5%* (`spherical_chebyshev`) but −2.0%* (`batched_morbo`) — a sign flip by solver (method dependence).

3.4 Penicillin fermentation (M=3, d=7)

A fed-batch bioreactor ODE integrated over the run; the raw response is the checkpointed state trajectory $g(x) = \{s(x, t_k)\}_k$, reduced to (yield, CO_2, time). **Result:** −2.5%* (`batched_morbo`) — a coupled trajectory of one process; composite hurts.

4 Power-flow benchmarks (MOBO-HighDim contribution)

Real IEEE 14-bus optimal power flow (CEC2021 RWCMOP). With complex bus voltages V (design variables, slack V_1=1) and sparse admittance $Y = G + jB$, one solve gives the per-bus complex power

$$I = YV, \qquad S = V \odot \bar{I}, \qquad P_{sp} = \Re(S), \quad Q_{sp} = \Im(S).$$

Because Y is *sparse*, each $P_{sp,k}$ depends only on bus k's neighbors — the source of decoupling. The raw response is $g(x) = (P_{sp}, Q_{sp}, \dots)$ and objectives are decoupled sums over it.

4.1 RCM40 (M=2, d=34)

$f_1 = \sum_k P_{sp,k}$ (active loss), $f_2 = -\Im(I_2) + \sum_{k\geq 2} Q_{sp,k}$ (reactive loss). **Result:** +0.8%* (`batched_morbo`).

4.2 RCM46 (M=4, d=34)

Adds fuel cost $f_1 = \sum_k (b_k P_{g,k} + c_k P_{g,k}^2)$ and voltage deviation $f_4 = \sum_{k\geq 2}(1 - |V_k|)^2$ to RCM40's two loss objectives. **Result:** +5.4%* (`batched_morbo`), +19.0%* (`spherical_chebyshev`). The win grows from RCM40's 2 to RCM46's 4 decoupled objectives.

4.3 MOOPF-5 (M=5, d=34; constructed)

RCM46's four objectives plus the Kessel–Glavitsch L-index voltage-stability margin. Partitioning Y into generator (PV/slack) and load blocks and forming $F = -Y_{LL}^{-1} Y_{LG}$,

$$L_j = \left| 1 - \sum_{i \in \text{gen}} F_{ji} \tfrac{V_i}{V_j} \right|, \qquad f_5 = \max_{j \in \text{load}} L_j.$$

$g(x)$ appends the per-load-bus L_j to RCM46's response. **Result:** +23.5%* (`batched_morbo`, n=20), +14.6%* (`spherical_chebyshev`) — the largest win in the suite. (Not a CEC2021 case; a constructed benchmark to test whether more decoupled objectives amplify the advantage.)

5 Controlled decoupling benchmark

5.1 Blocked DTLZ2 (M=2, d=49)

DTLZ2 with the objective held fixed while the distance term is exposed as $K = \lceil (d-1)/b \rceil$ block partial sums,

$$\text{block}_i = \sum_{j \in \text{block } i} (x_j - 0.5)^2, \qquad \sum_i \text{block}_i = g \quad \forall b.$$

The block size b dials the effective input-dimension of each raw component (b/d) from $\sim 1/d$ (decoupled) to 1 (coupled) with everything else fixed. **Result:** composite advantage falls monotonically from +11.9% (b=1) to +7.4% (b=48), Pearson $r = -0.997$ (`ablation_results/_decoupling_sweep/`). Causal confirmation that decoupling drives the advantage.

6 Results summary

Benchmark	Best pair	Dim/Obj	n	ΔHV
Power flow				
MOOPF-5	batched_morbo	34/5	20	+23.5%*
MOOPF-5	spherical	34/5	13	+14.6%*
RCM46	spherical	34/4	18	+19.0%*
RCM46	batched_morbo	34/4	20	+5.4%*
RCM40	batched_morbo	34/2	20	+0.8%*
Synthetic				
Langermann–Ackley	standard	6/2	20	+8.0%*
DTLZ2	batched_morbo	6/2	20	+5.3%*
Five-Ackley	batched_morbo	6/5	20	+5.0%*
Ackley–Griewank	standard	6/2	20	+2.7%*
DTLZ2	batched_morbo	100/2	20	+0.3%*
DTLZ2 (5obj)	batched_morbo	6/5	20	+0.5%*
Projected Langermann	spherical	500/2	8	+0.5%*
DTLZ2	batched_morbo	600/2	20	+0.1%
Ackley–Griewank	batched_morbo	50/2	20	+0.1%
Real-world (chem / optics / medical / bio)				
CORT TG119	spherical	418/3	20	+1.5%*
CORT TG119	batched_morbo	418/3	20	−2.0%*
Summit SnAr	batched_morbo	4/2	20	+0.3%*
Nanoparticle RGB	standard	6/3	20	+0.2%
Penicillin	batched_morbo	7/3	20	−2.5%*

Table 1: Composite-minus-direct final-hypervolume change, best/most-complete solver pair per benchmark.

7 Key findings

1. **Wins scale with decoupled-objective count** on the identical IEEE 14-bus system: RCM40 (+0.8%) → RCM46 (+5.4%) → MOOPF-5 (+23.5%).
2. **Method dependence:** CORT flips sign between `batched_morbo` (−2.0%) and `spherical_chebyshev` (+1.5%); the advantage is strongest with trust-region acquisition.
3. **Chemistry / trajectory / global-solve benchmarks do not benefit** (SnAr, Penicillin, CORT-on-morbo): the raw response is one globally-coupled state, no easier to model than the objective.
4. **Decoupling, not reduction nonlinearity, is the discriminator:** RCM46 wins with a linear reduction while SnAr's nonlinear E-factor ratio ties. The effective input-dimension of g separates wins from losses.
5. **Causal confirmation (blocked DTLZ2):** varying only the raw-response effective dimension moves the advantage monotonically, $r = -0.997$.